

Module 11

AI-Assisted Design — Tier 3: Advanced

1. Text-to-3D Generation

Newer AI tools can generate a rough 3D model directly from a text description or a reference image — useful for quick concepting, though outputs typically need cleanup before they're truly print-ready.

2. AI Inside CAD Tools

- Auto-generating variations of a design (generative design)
- AI-assisted retopology and mesh cleanup for imported/scanned models
- Smart suggestions for wall thickness, supports, and structural weak points

3. Realistic Expectations

AI-generated 3D models today are best treated as a fast **starting point**, not a finished file — expect to clean up geometry, fix non-manifold errors, and check wall thickness manually before printing.

4. Practical Workflow

- Generate a base concept with an AI tool
- Import into CAD/mesh-repair software (e.g. to fix errors)
- Verify printability (walls, overhangs) before slicing

End of Module 11 — Next: Module 12, AI in Slicing & Print Optimization